one had produced a machine capable of a speed of even 200 MHz. All these systems were stepping stones for further development in the supercomputer arena.

What's the difference?

Supercomputers do not differ from desktop PCs in processing power, storage capacity and memory alone. It's the architectural and algorithmic change, which defines a supercomputer. There is a difference in the manner in which a desktop computer and a supercomputer process data—supercomputers carry out large amounts of parallel and vector data processing. These machines are built on multi-processor architectures with unified memory of a high order wherein computations take place in terms of megaflops, teraflops or petaflops. They work on special algorithms programmed into their hardware and usually work with more than 10-15 processors with memory architecture comparable to a desktop PC's processing power. The parallel processing of information allows these computers to analyse and do complex arithmetic and mathematical calculations and predictions. Hence, they are employed in areas such as weather prediction, DNA computations, Artificial Intelligence, molecular modelling of drugs, space research, statistical analysis and more.

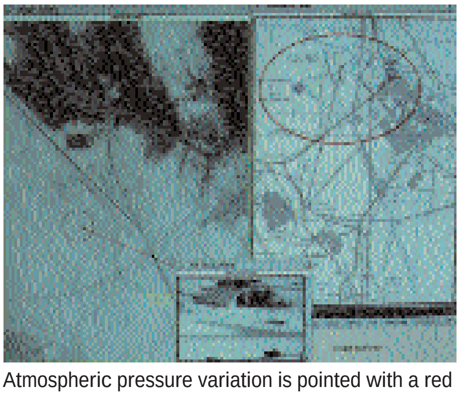
The supercomputer can either be a single computer or a network of nodes or a grid. These computers mainly rely on the parallel and vector processing techniques that are employed in the making of these machines. The algorithms that are used to parallel process data could either be hardwired or it could just be the software which makes effective use of the hardware and taps the power of a supercomputer's processors and their unified memory.

The network and the grid story

To form a supercomputer, many smaller computers can be networked together to work in unison on a single process and can be controlled by software which can carry out effective data management within this network.

Grid computing is an entirely new concept. A grid computer is defined as a whole spectrum of clustered and distributed data processing techniques that allow many machines to be ganged up together to solve complex computational problems. Its principle lies in utilising the idle time of computers connected to the Internet. SETI@home (Search for Extra Terrestrial Intelligence), for instance, is a project, which lets you donate CPU cycles to search for extra terrestrial life. The idea here is to harness the idle time of computers connected to the Internet and use them to perform complex arithmetic, which would otherwise require large parallel processing supercomputers costing millions of dollars. On an average, a desktop PC uses 80 per cent of the CPU power and time and rest of the time it remains idle.

United Devices is an initiative that saw its birth from the SETI@home project, which is being considered as the largest and most successful distributed computing project in history. United Devices helps organisations all over the world carry out their projects on virtual supercomputers created with the power of all the connected computers.



Atmospheric pressure variation is pointed with a red circle. Simulated with 22 supercomputers

A Super-network

Although the idea of creating supercomputing power from PCs connected to the Internet is certainly good, certain operations (especially in research) do require the networking of true supercomputers—those with parallel and vector processing capabilities. The applications that run on these grids need more bandwidth than that offered by a local area network or the Internet. More importantly, these systems handle extremely sensitive information, which require more robust security. Researchers and programmers worldwide have collaborated on a project that answers these supercomputing needs. The project, code named Globus, is an open source initiative for grid computing and is focused at major corporations that can use grid comput-

Vector Processing

A vector, in computer terms, comprises the starting address in the memory, along with the length through which the vector data runs and the length of each element in the array, called stride. Machines with vector processing capabilities read the entire vector or the array into the memory and perform operations on them in a single cycle and write them back to the memory also in just a cycle.

Basically vector processing is a SIMD (Single Instruction Multiple Data) parallel processing technique. A scalar technique would require that one instruction at a time for each data be operated on. Vector instructions take one or more vector registers (a place where the computer stores the data it operates on) as operands and perform the same operation on each related element to provide the resultant vector. Many scalar instructions clubbed together effectively form a single vector instruction. Here, the same instruction is carried out on all the vector elements. You can execute independent vector instructions in parallel to quicken computation and to decrease the turn around time for a job. This method of processing is effectively used in supercomputers to make them work simultaneously on an array of data at a faster rate.